



CYBERPUNK C64 COMMAND

User manual for the Mac

A breadbin with a network socket is a wonderfully absurd thing. This app makes the most of it: it inserts disks, starts programs, digs through your SID archive and turns up the case lighting — all from your desk, while the C64 on the table beside you does exactly what it could already do in 1982.

This manual covers everything the app can do, in the order you usually need it. You do not have to read it front to back. If you already know what you are looking for, jump to the chapter.

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1 — What you need

The list is short, and most of it is probably already sitting on your desk.

- A **Mac** with Apple silicon and macOS 13 or later.
- A **C64 Ultimate** or an Ultimate 64 board, on the same network as the Mac. The REST interface has to be switched on in the firmware.
- The **IP address** of your Ultimate. You will find it in the Ultimate menu under the network settings.
- Optional: disk images on the Mac, an unpacked **High Voltage SID Collection** and a good mood.

Without an Ultimate the Disk Workbench still works in full — opening, inspecting, editing and building images all happens locally on the Mac. The other modules need the connection.

Electron and Node.js are included in the app; neither needs a separate installation.

Installing

- 1 Drag the app out of the archive into your **Applications** folder.
- 2 The very first time, right-click the icon and choose **Open**. macOS asks once; after that the app starts with a double-click like any other.

WHY THE DETOUR THE FIRST TIME?

The app is not registered with Apple. Right-clicking and choosing Open is the official way to confirm once that you really do want to run it. After that you will not be asked again.

2 — First start

On first launch the startup screen stays for about three seconds. A one-time setup then takes you through four short steps.

Four steps to get started

- 1 Choose your **language**: German or English.
- 2 Try the **animations** and keep them on or switch them off.
- 3 Enter the **Ultimate address**, for example 192.168.1.64. Leave it empty if you have no device yet.
- 4 Choose the **start view**: the home screen or an available module.

With an address entered, the app then tries to connect. For an Ultimate password or later changes, open the gear in the top right and choose **Save & connect**. The Disk Workbench also works without an Ultimate.

Connecting the File Manager

The File Manager (F3) talks to the Ultimate directly over FTP. The connection is built into the app — you need neither a helper program nor an open terminal window.

- 1 Under **File Manager · FTP**, enter the user name. This is usually `anonymous`.
- 2 If the FTP account has a password, enter it in the field below.
- 3 Click **Save & connect**. When you open **F3**, the File Manager connects by itself.

NO SECOND PROGRAM NEEDED

FTP uses the same Ultimate address as the other modules. The address and user name stay saved; passwords last only for the current app session.

3 — Finding your way around

The app is split into modules, and each one has a function key. You can see them all in the bar along the bottom — which is both the navigation and the reminder of which key goes where.

The home screen

If you chose the home screen as your start view, its tiles show the enabled modules. A click opens a module. Modules that need the Ultimate are greyed out without a connection. You can choose another start view during setup or later in the settings.

The bar at the bottom

The bar shows every switched-on module with its key. **Info** always sits at the far right and always carries the highest number — switch on another module and Info moves one key further rather than taking its number away.

The status line

The very bottom says what the app is doing, or did last. For longer jobs — mounting an image, reading a folder — a progress indicator runs there. To its right you can see whether the Ultimate is reachable.

The gear

Top right, the gear opens the settings: connection, language, start view and the module switches. Chapter 12 goes through them.

4 — F2: The Disk Workbench

The heart of the thing when disks are involved. Look inside a D64, D71 or D81, take files out, throw files away and build yourself new images.

Opening an image

Click **Open disk image** and pick a file from the Mac. The app reads it and shows the directory on the left, the way the C64 would show it — disk name, ID and the number of free blocks.

If the image lives on the Ultimate, open it from the File Manager (F3) instead: the route is shorter, and the app remembers where the image came from.

Browsing the directory

A click selects an entry, the arrow keys walk the list. A preview appears on the right straight away, and it follows the file type:

- **Pictures** — Koala, Doodle, Hires and similar formats are drawn directly.
- **SID files** — title, composer, year and the number of subtunes.
- **BASIC programs** — as a readable listing, not as a wall of numbers.
- **Everything else** — as a hex dump, for a look at the raw bytes.

Empty entries with zero blocks — often just decoration in the directory — are shown dimmed, and the app offers no way to start them. There is nothing in them to start.

What you can do with a file

Every action for the selected entry lives on the right in the details panel, in a fixed order: starting first, then the tools, then managing the entry at the bottom.

Action	What happens
Mount and start	Inserts the image in the Ultimate and starts the program. The fast route.
Start via LOAD	Types <code>LOAD"*",8,1</code> and <code>RUN</code> on the C64. For demos with their own loader.
Play	Appears for SID files and plays the tune on the Ultimate.
Add to basket	Puts the file aside for a new image.
Extract	Saves the single file to the Mac.
Rename	Changes the name inside the image. D64 only.
Delete	Removes the file from the image. D64 only.

The filter

Above the list, the filter hides individual file types. Useful on a full disk when you only want to see the PRGs and all those SEQ files are in the way.

The BAM view

The Block Availability Map shows as a grid which sectors are taken and which are free. Every square is a sector, every row a track. It is the quickest way to see how full a disk really is — and how a file is spread across the tracks.

40 TRACKS ARE RECOGNISED

Some images use tracks 36 to 40, which a real 1541 knows nothing about — formatted with SPEED DOS or DOLPHIN DOS, for instance. The app recognises both, shows the extra tracks in the grid and counts their free blocks. For a format it cannot read with confidence it draws those tracks dashed and refuses to delete, rather than damaging a disk.

The collection basket

The basket is a shelf that survives across images. Open a disk, put two files in it, open the next one, add another — then build one new image out of all of it. PRG files from the Mac and tunes from the SID Collection land here too.

Once everything is together, write the contents into a new or an existing D64. **Download image** asks for a name and saves it on the Mac.

5 — Starting programs: the two routes

The app offers two ways to start a program. They reach the same destination but take very different roads — which is exactly why the second one helps when the first fails on a demo.

Mount and start

The app inserts the image, takes the program out of it and pushes it straight into the memory of the C64. That is fast and works for the vast majority of programs.

Start via LOAD

Here the app does what you would otherwise type yourself: it inserts the image, resets the C64, waits for BASIC to come up and then types `LOAD"*",8,1` followed by `RUN` — character by character into the keyboard buffer, as if you were sitting there.

It is slower but more honest: the program really is loaded from the disk, with everything that entails. Demos with their own fast loader need exactly that, because their loader wants to reach the disk itself.

THE APP REMEMBERS WHAT WORKED

Whichever route worked for an image is remembered. Next time the app takes the right one straight away. You can always pick the other one anyway.

A word on patience: after inserting the image the app waits a few seconds before resetting the C64. Inserting throws the drive off its stride for a moment, and a reset landing in the middle of that helps nobody. After that, loading takes as long as loading from a real disk takes.

SID files

SIDs are not mounted. The app recognises them by their header, offers **Play** and hands the tune straight to the music player in the Ultimate.

6 — F3: The File Manager

A look at the SD card in the Ultimate — without taking it out.

On the left you see the folders and files on the Ultimate, on the right a preview of the selected file. A double-click on a folder goes in; the path line at the top takes you back.

What you can do here

- **Upload** — put files from the Mac onto the Ultimate.
- **Download** — fetch files from the Ultimate to the Mac.
- **Rename** and **Delete** — straight on the card.
- **Create folder** — for order in the archive.
- **Open image** — loads a D64, D71 or D81 into the workbench.
- **Start image** — inserts it in the drive right away.

For PRG, CRT and SID files the preview offers the matching launch action directly.

IMAGES ON THE ULTIMATE ARE NOT COPIED

When you insert an image that already sits on the card, the app just points at it instead of shifting it back and forth. That is not only faster: for images with 40 tracks it is the only route that leaves the extra tracks intact.

The app remembers which folder you were last in and which file was selected. Switch between F2 and F3 and you come back to the same place.

7 — F4: C64 Remote Control

Everything you would otherwise have to get up and press buttons for.

Area	What you find
Drives A and B	Insert an image, eject, reset the drive, switch it on and off.
C64 control	Reset, reboot, pause and continue, menu button.
System	The Ultimate's heavier commands — to be used with care.
Configuration	Store changes permanently in the Ultimate.

Insert image transfers whatever image is currently open in the workbench as a read-only test copy. For real work with a disk, go through the File Manager instead.

DO NOT FORGET TO SAVE PERMANENTLY

Changes to the Ultimate configuration take effect at once but are gone after a restart. Only **Save permanently** writes them down for good.

8 — F5: Media

The launch pad for single files. Pick a file from the Mac — or take the one currently selected in the workbench — and send it to the C64:

- **PRG** — loaded and started.
- **CRT** — inserted as a cartridge.
- **SID** — played.
- **MOD** — played, if your firmware supports it.

Handy for a quick look: try a program without building an image first.

9 — F6: The Light Studio

The part that does nothing a C64 of 1982 could do — and the one people show off most.

Case lighting and keyboard lighting are controlled separately: mode, pattern, brightness, colour and tint. Every change is visible immediately.

On top of that sit **30 ready-made light shows** in three banks of ten. One click and both lights change together. The show you picked last stays marked, even after you leave the module.

ONE AT A TIME

While a setting is on its way to the Ultimate, both cards are briefly locked. That is deliberate: two changes at once confuse the lighting. If something fails, the app reads the real state back from the device — what you see on screen is then what is actually set, not what was supposed to be set.

10 — F7: The SID Collection

Over 60,000 tunes, and you are looking for one. This module turns that into a matter of seconds — and plays what you find on real silicon.

Setting it up once

The module works with the **High Voltage SID Collection**, which you download from hvsc.c64.org. Unpack it anywhere on the Mac — inside, the folder is usually called `C64Music`.

- 1 Open F7 and click **Choose HVSC folder**.
- 2 Pick the folder you unpacked.
- 3 Wait a moment while the app takes stock of the collection.

After that the top line says how many tunes were found and when the index was made. It is kept, so everything is there immediately at the next start. If you have updated the collection, **Read again** picks up the changes.

WHAT THE APP KEEPS — AND WHAT IT DOES NOT

Only paths, titles and composers are stored, not the music files themselves. The index is a list, not a second archive. Nothing is put on the Ultimate: to play a tune, that one file goes across, a few kilobytes, and that is it.

Searching

Just start typing. **Every word has to match** — so "hubbard commando" shows only what hits both, instead of inflating the result list. Upper and lower case make no difference, and neither do the underscores in the file names.

The exact title comes before the variation: for "commando", *Commando* ranks above *Commando Remix*. The buttons underneath narrow things down to musicians, games or demos.

Listening

A click selects a tune, a double-click plays it right away. On the right you get what the file header says: format, SID chip, clock and size. If a tune has several subtunes you can choose which one plays before you start it.

Whatever you like, put aside with **Add to basket** — and build a disk from it later in the workbench.

Giving stars

Under the title sit five stars. Click the third and the tune has three. Click the same star again and the rating is gone — no separate clear button needed.

Rated tunes rise to the top of the result list and show their stars there. The **Rated** button hides everything else — turning the huge archive into your own shortlist.

YOUR RATINGS SURVIVE A MOVE

What is stored is the path inside the collection, not the full one. Move the HVSC folder to another disk or read it in again, and your stars stay where they are.

11 — Changing the look

If the yellow on black is too loud for you, bend it into shape. This goes further than you might expect.

The design editor opens through **Appearance** → **Design** or with **Cmd+Shift+D**. It appears in a window of its own beside the app, so you see every change at full size straight away.

Colours, type, spacing

Thirty-one settings in four groups: colours, typography, spacing, and frames and motion. Every change shows in the app at once — but nothing is saved until you say so. So you can experiment in peace.

For text colours the editor checks the contrast against the background. If a combination becomes hard to read, the row turns red — the warning stops you from nothing, it just tells you in time.

Four ready-made looks

- **Original** — yellow on black, as shipped.
- **C64 blue** — the familiar colour of the breadbin.
- **Green monitor** — for friends of phosphor.
- **Amber** — its warmer brother.

Arrangement and density

The order of the modules can be changed with the arrows — this shows in the bar at the bottom and on the home screen. Density squeezes or stretches all spacing at once: **Compact** fits more on screen, **Spacious** lets it breathe.

Through the three-dot menu you can save your design as a file and load it back — handy for sharing it or carrying it between machines.

12 — Settings and modules

There is more behind the gear in the top right than just the connection.

Setting	What for
Ultimate features	Switches off every module that needs the device. Without an Ultimate the workbench stays usable.
Ultimate connection	IP address and, if needed, the password.
File Manager · FTP	User name and, if needed, the FTP password. The Ultimate address applies here too.
Language	German or English, changed instantly.
Start view	Which module appears when you open the app.
Modules	Every module can be switched on and off individually.

A module that is switched off disappears completely: from the bar, from the home screen, from the start view choices and from the keyboard shortcuts. If light and media are not your thing, switch them off and you have a tidier app.

13 — All keyboard shortcuts

Key	Effect
F1	Home screen
F2	Disk Workbench
F3	File Manager
F4	C64 Remote Control
F5	Media
F6	Light Studio
F7	SID Collection
F8	Info and settings
Cmd+Shift+D	Open the design editor
Arrow keys	Walk through lists
Enter	Start what is selected
Double-click	The same with the mouse
Esc	Close a dialog

The module numbers follow the list — switch one off or change the order and they shift accordingly. Info always keeps the last number.

14 — When something goes wrong

The app cannot find the Ultimate

- Are the Mac and the Ultimate on the same network? A guest network often keeps devices apart.
- Is the IP still right? Many routers hand out a different one after a restart. Check the Ultimate menu.
- Is the REST interface switched on in the firmware?
- If your Ultimate has a password set, the app needs it too.

The File Manager will not connect

First check that the Ultimate address is correct and that the Mac and Ultimate are on the same network. Then open the gear: under **File Manager** · **FTP**, the user name and, if needed, FTP password must be correct. Click **Save & connect**, then switch away from F3 and back.

A demo will not start

Try **Start via LOAD**. Demos with their own fast loader really do want to load from the disk, and the direct route into memory skips exactly that. If loading takes noticeably long: that is normal — a real disk is a real disk.

A disk looks strange

Images with an unusual number of tracks or their own file system cannot always be read. The app says so and leaves deleting alone rather than guessing. Looking at files and taking them out usually still works.

The SID Collection barely finds anything

Then the folder you picked probably does not point at the full collection but at an update package or a subfolder. Make sure to choose the folder that contains `C64Music` — the top line says how many tunes were found. For the complete collection that is tens of thousands.

Something else

The status line at the bottom usually says what happened last. And if nothing works at all: quitting and reopening the app costs two seconds and clears up a surprising amount.

15 — Thanks and small print

This app would not exist without other people's work. Thanks go to:

- The **Ultimate** team for the hardware and the open interface that makes all of this possible.
- The curators of the **High Voltage SID Collection**, who have held together a music archive without equal for decades.
- Style, for the **C64 Pro Mono** typeface that gives this app its face.
- Everyone still writing demos, games and music for a home computer from 1982.

The app ships no third-party programs, no disk images and no music. Whatever you open with it belongs to you — or to whoever made it. The High Voltage SID Collection has terms of its own; they are on hvsc.c64.org and take a minute to read.

And one more thing no manual can really write down: the best part of all this is not the interface. It is the moment the screen next to you flashes blue, the drive starts rattling and something gets going that somebody wrote forty years ago.

READY.

